

Literature Benchmark Brief: Bossanyi_2000_WindEnergy

Title: The design of closed loop controllers for wind turbines

Authors: E. A. Bossanyi (2000)

Venue: Wind Energy | **DOI:** 10.1002/1095-4244(200007/09)3:3<149::AID-WE34>3.0.CO;2-H

Cluster: Cluster C: Aerodynamics & Control Cliff

Core Theoretical Mechanism & Model Formulation

Gain scheduling PI pitch controller design; sensitivity inversion near rated power

Empirical Scale & Experimental Dataset

Aeroelastic simulation and full-scale field turbines

Relevance to IEEE TSTE & Sustainable Energy Systems

Explains why turbine power response becomes non-linear near rated wind speed

Identified Gaps & Operational Vulnerabilities

Assumes instantaneous sensor feedback without communication network delay

Role in Manuscript Defense & Reviewer #2 Proof

Explains the physical origin of the control cliff: pitch controller saturates aerodynamic lift